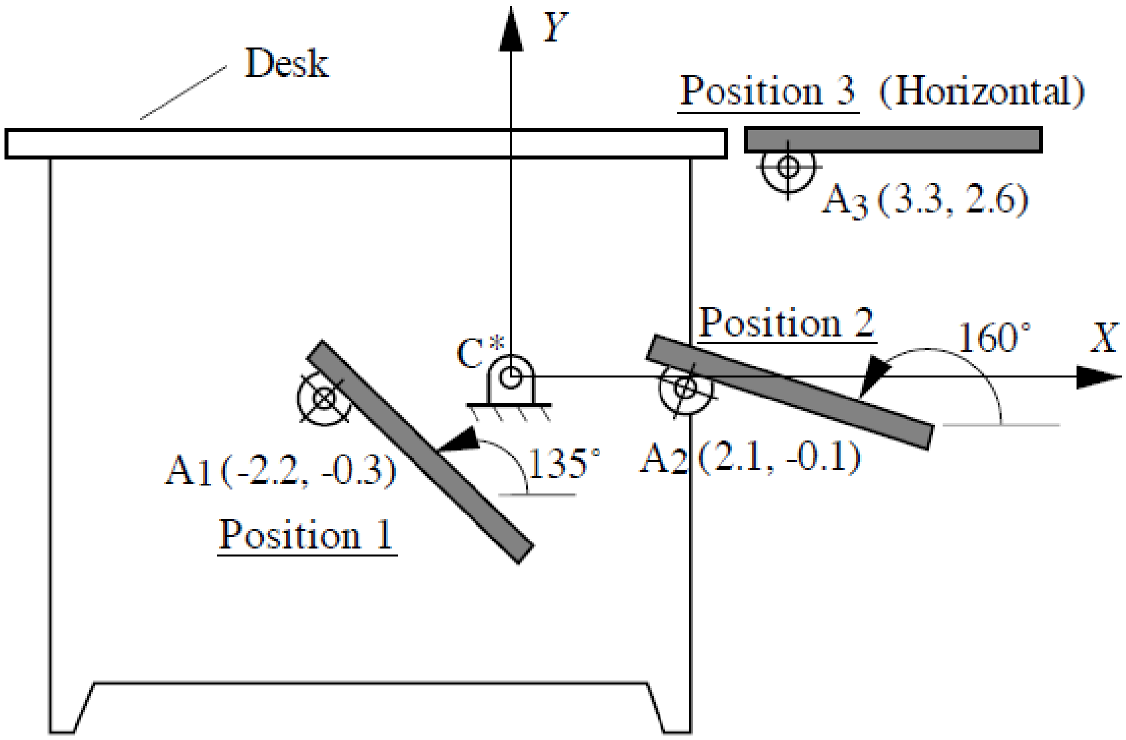

1.

A mechanism must be designed to move a computer terminal from under the desk to top level. The system will be guided by a linkage, and the use of a four-bar linkage will be tried first. As a first attempt at the design, do the following:

a) Use C^* as a center point and find the corresponding circle point C in position 1.

b) Use A as a circle point and find the corresponding center point A^*.

c) Draw the linkage in position 1.

d) Determine the type of linkage (crank rocker, double rocker, etc.) resulting.

2. Graphically synthesize a slider crank mechanism with a stroke length of 5 cm and offset of 2 cm that will have a quick return ratio $= \dfrac{\theta_2^{12}}{2\pi - \theta_2^{12}} = 3$ (all angles in radians). Draw the synthesized mechanism clearly in the two dead center configurations

3.

Determine the link lengths and draw a four-bar linkage that will generate the function $\phi = \theta^2$ (θ and ϕ both in radians) for values of θ between 0.5 and 1.0 radians. Use Chebyshev spacing with three position points. The base length of the linkage must be 2 cm.

4.

Draw the displacement schedule for a follower that rises through a total displacement of 1.5 inches with constant acceleration for 1/4th revolution, constant velocity for 1/8th revolution, and constant deceleration for 1/4th revolution of the cam. The cam then dwells for 1/8th revolution, and returns with simple harmonic motion in 1/4th revolution of the cam.

The first part of the displacement schedule/profile will require a polynomial fit . After determining the same, find the expressions for minimum follower length and minimum base circle radius for each segment if a radial flat faced follower is to be used.

5.

In the figure given, axis y-y is fixed while axes x-x and z-z move with the arm. Gear 7 is fixed to the carrier. Gears 3 and 4, 5 and 6, and 8 and 9 are fixed together, respectively. Gears 3 and 4 move with planetary motion. If the tooth numbers are $N_2 = 16$T, $N_3 = 20$T, $N_4 = 22$T, $N_5 = 14$T, $N_6 = 15$T, $N_7 = 36$T, $N_8 = 20$T, $N_9 = 41$T, and $N_{10} = 97$T, determine the speed and direction of the output shaft.